



Predicting future outcomes analysis for Turtle Games

George Breese

Who are Turtle Games



Turtle Games are a game manufacturer and retailer with a global customer base. Turtle Games sells its own products, as well as other companies' products. Turtle Games' product range includes books, board games, video games and toys.



Aims of the analysis

Turtle Games are aiming to improve overall sales performance by analysing and considering customer behaviour trends.

- How can customers be segmented into groups, and which groups can be targeted by the marketing department?
- How do customers engage with, and accumulate, loyalty points?
- How can text data be used to inform marketing campaigns and make improvements to the business?
- Can we use descriptive statistics to provide insights into the suitability of the loyalty points data to create predictive models?



What are loyalty points?

- Loyalty points are a form of reward scheme that recognises loyal behaviours
- The loyalty points can be redeemed for rewards
- Easy for customers to understand
- Flexible to issue and redeem



Lance Walker, former CEO of Loyalty New Zealand writing on LinkedIn.



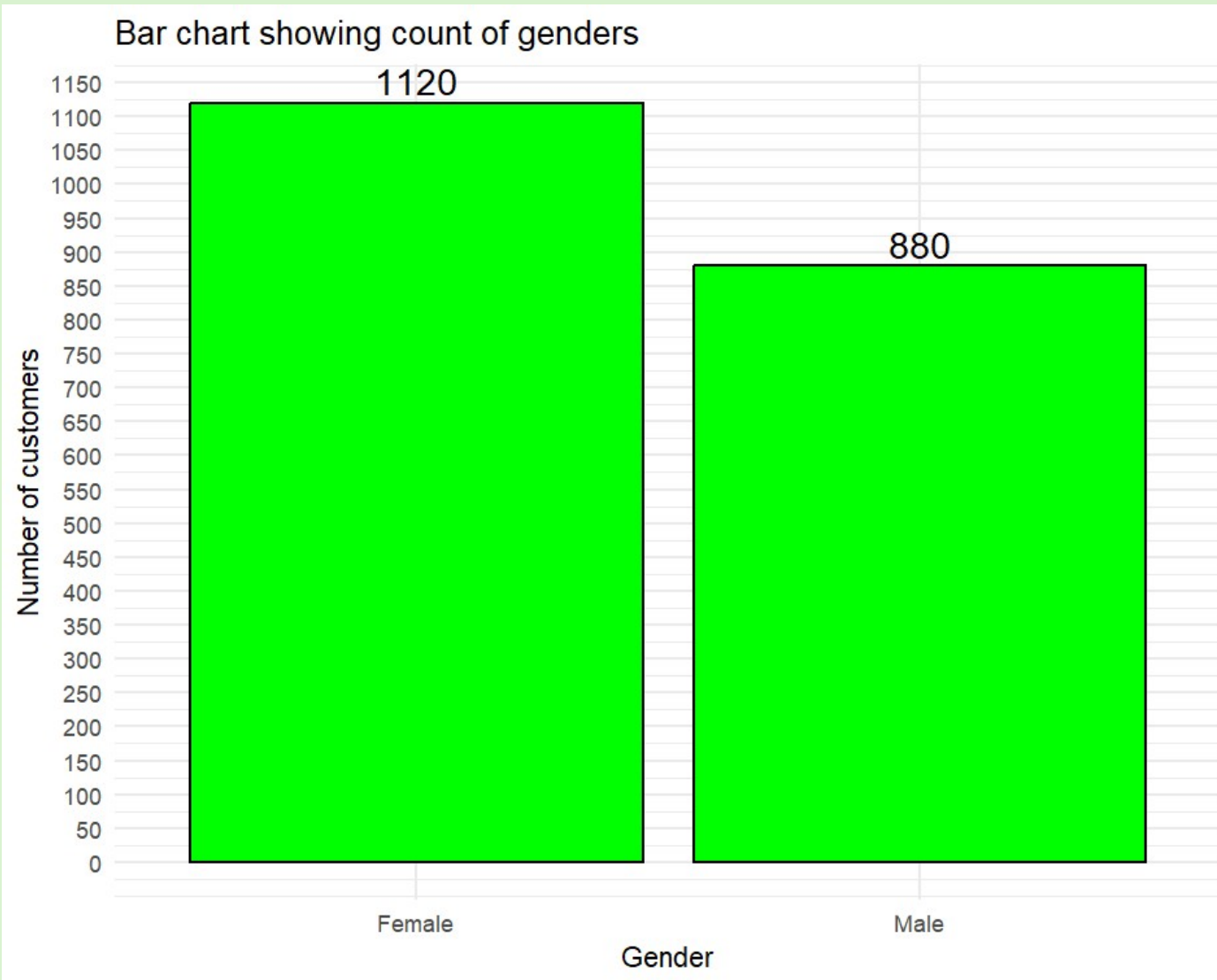
Exploring the data

There are 13 columns, 2000 rows

Columns include gender, age, loyalty points, education, income, spending score

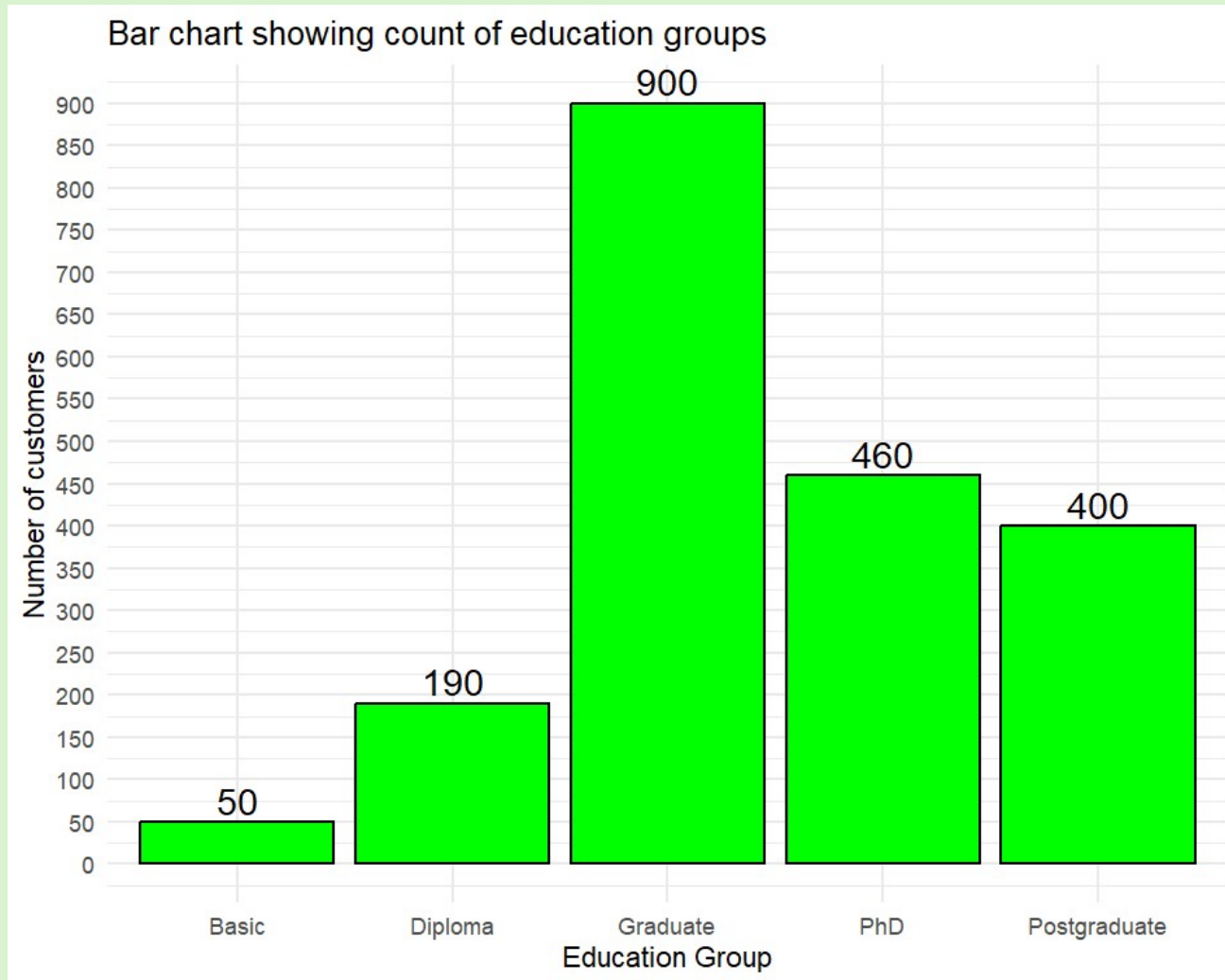


There are
more
females
than
males



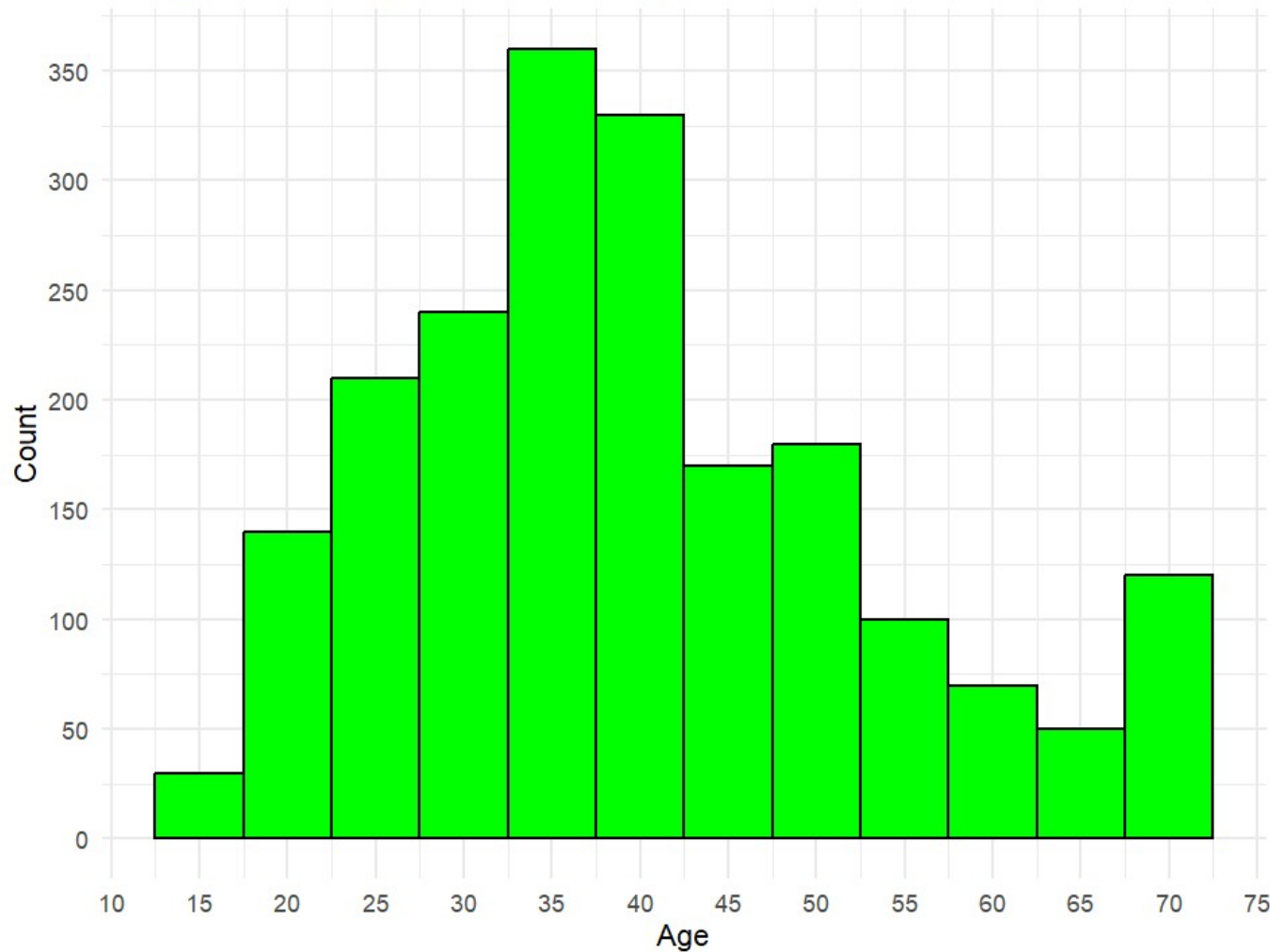


Most of the
customers
are
graduates





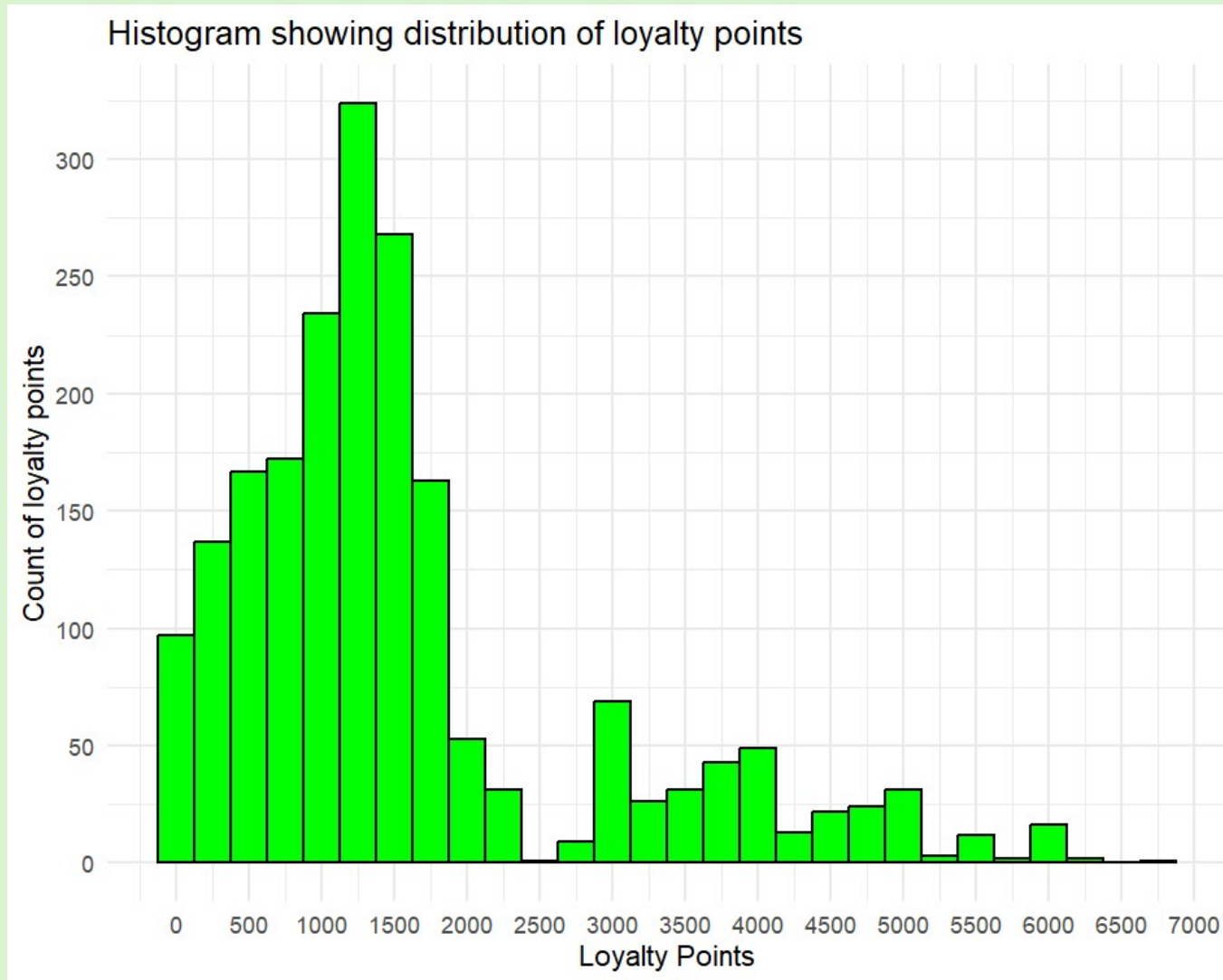
Histogram showing distribution of age

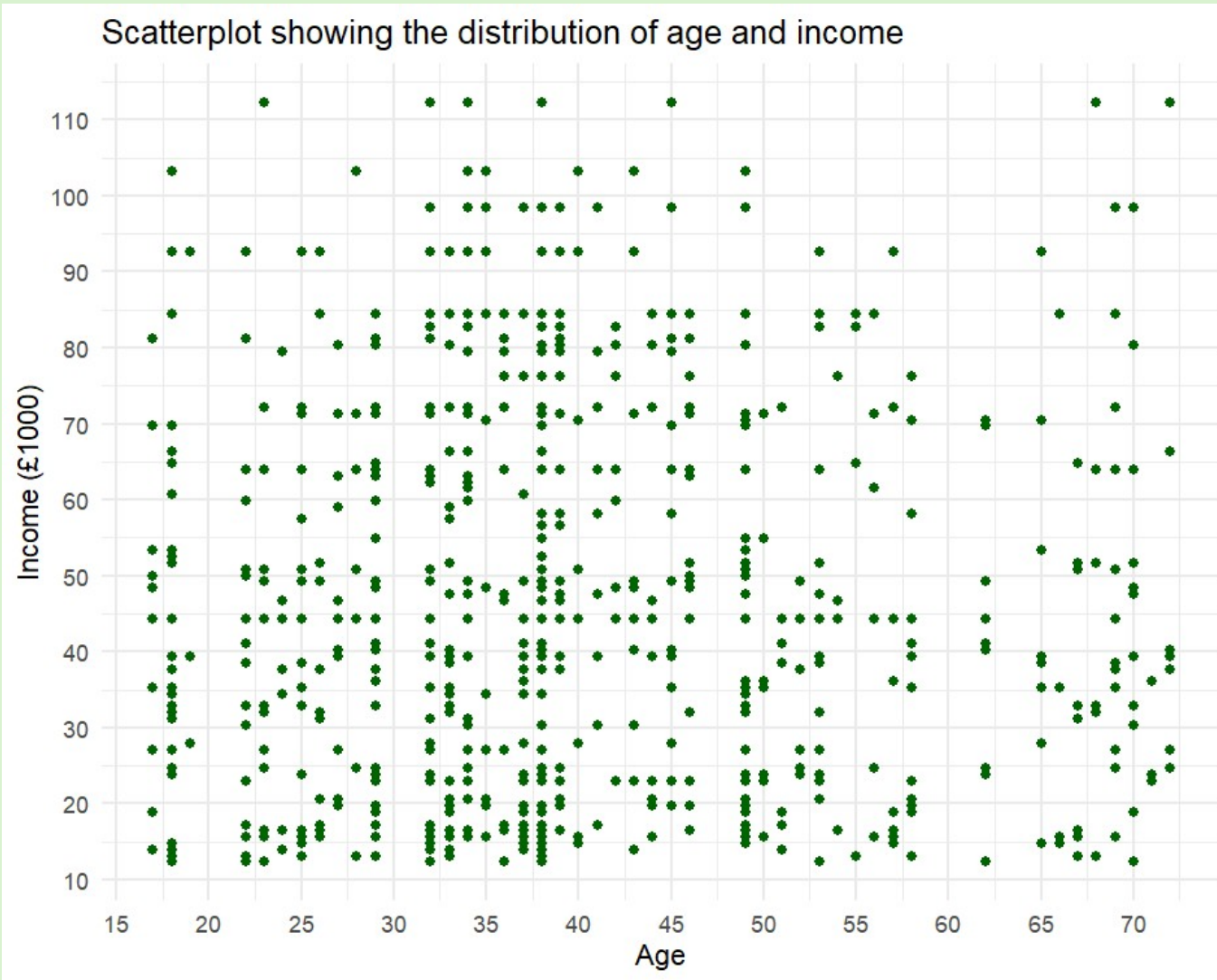


Most of the customers are younger than 45



Most of
the
customers
have
between
1000 to
1500
loyalty
points



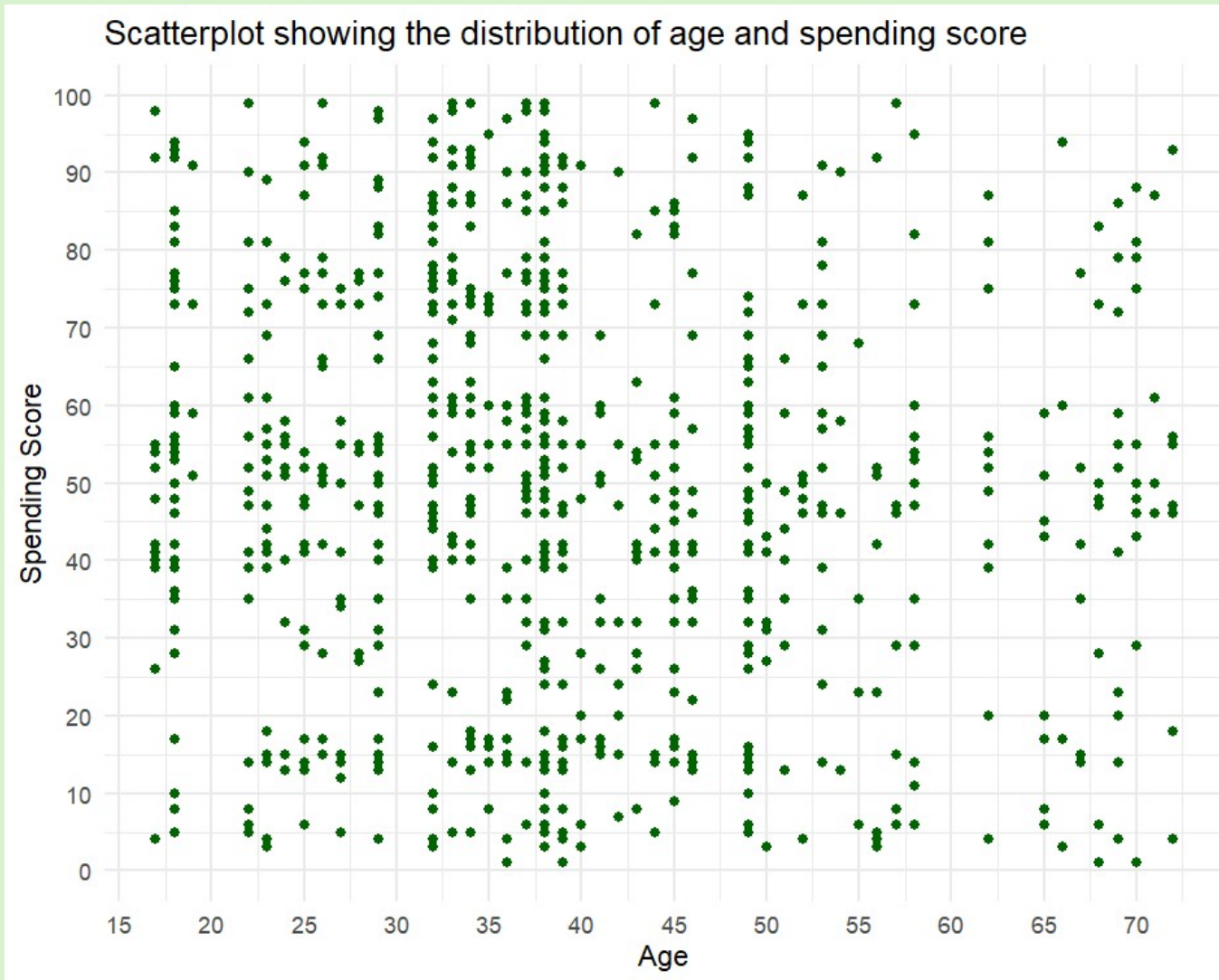


Each dot
here
represents
an age and
an income

It is very
varied

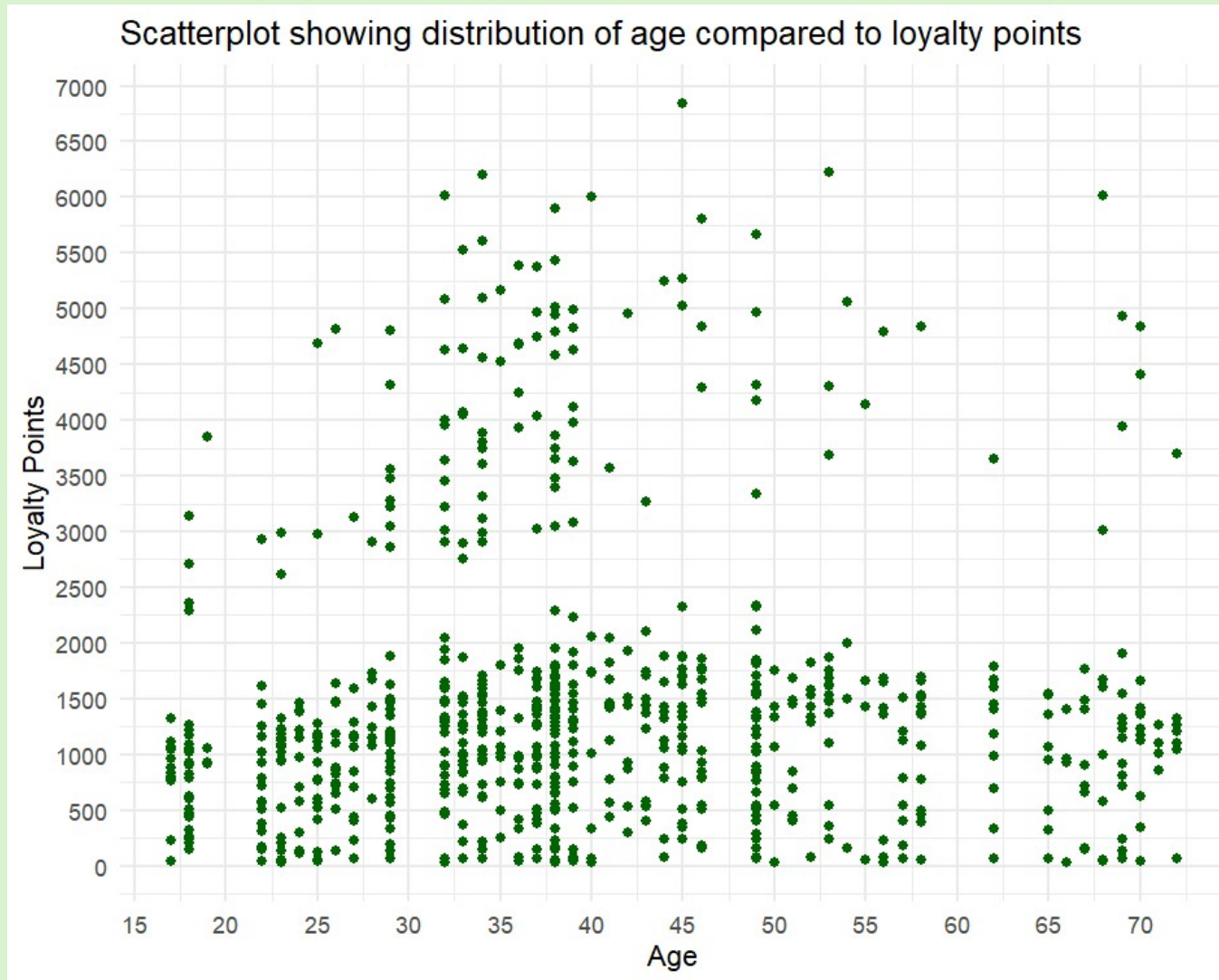


Very
varied
when we
look at
spending
score and
age



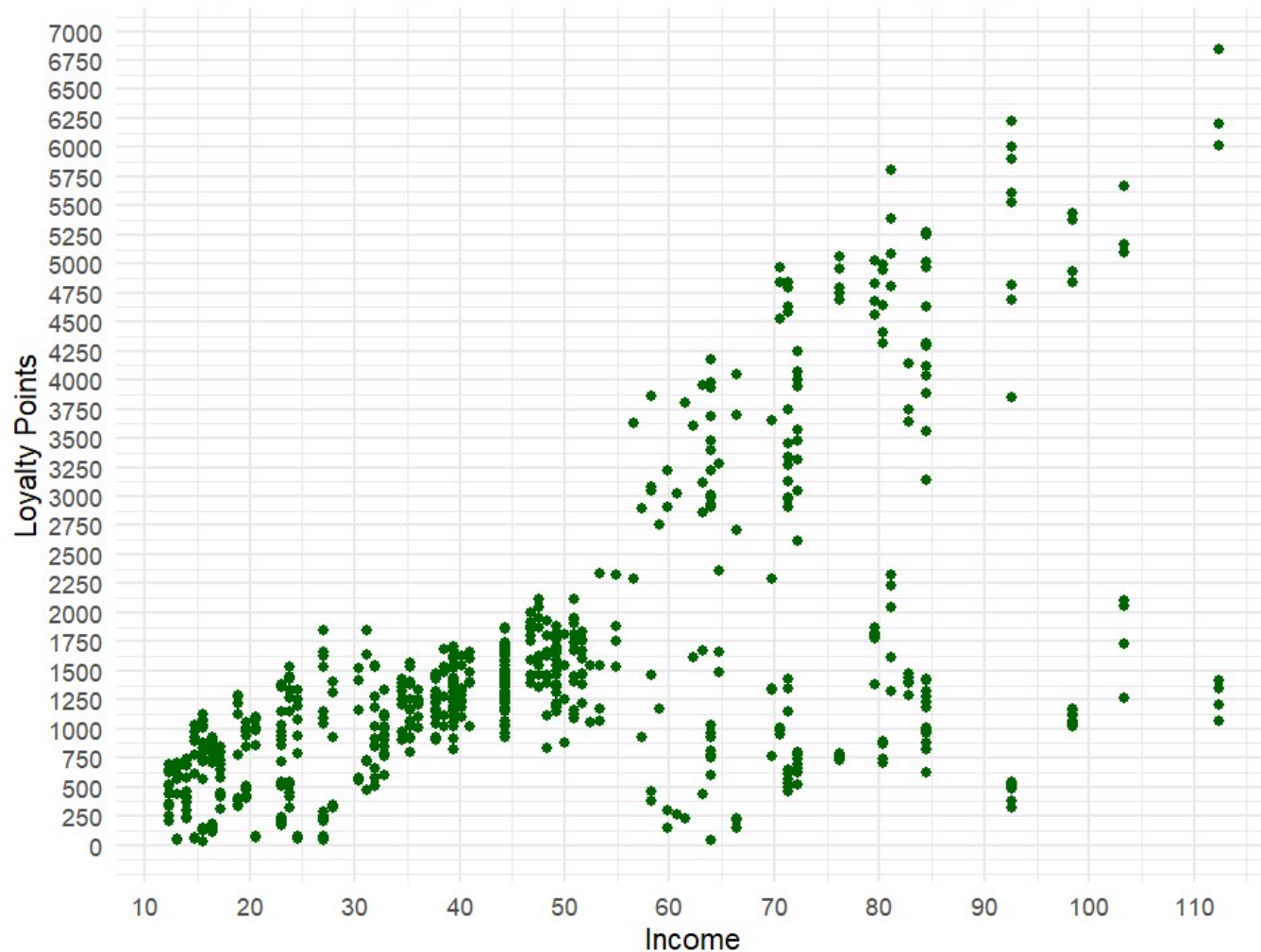


Most of the customers have loyalty points below 2000 regardless of age





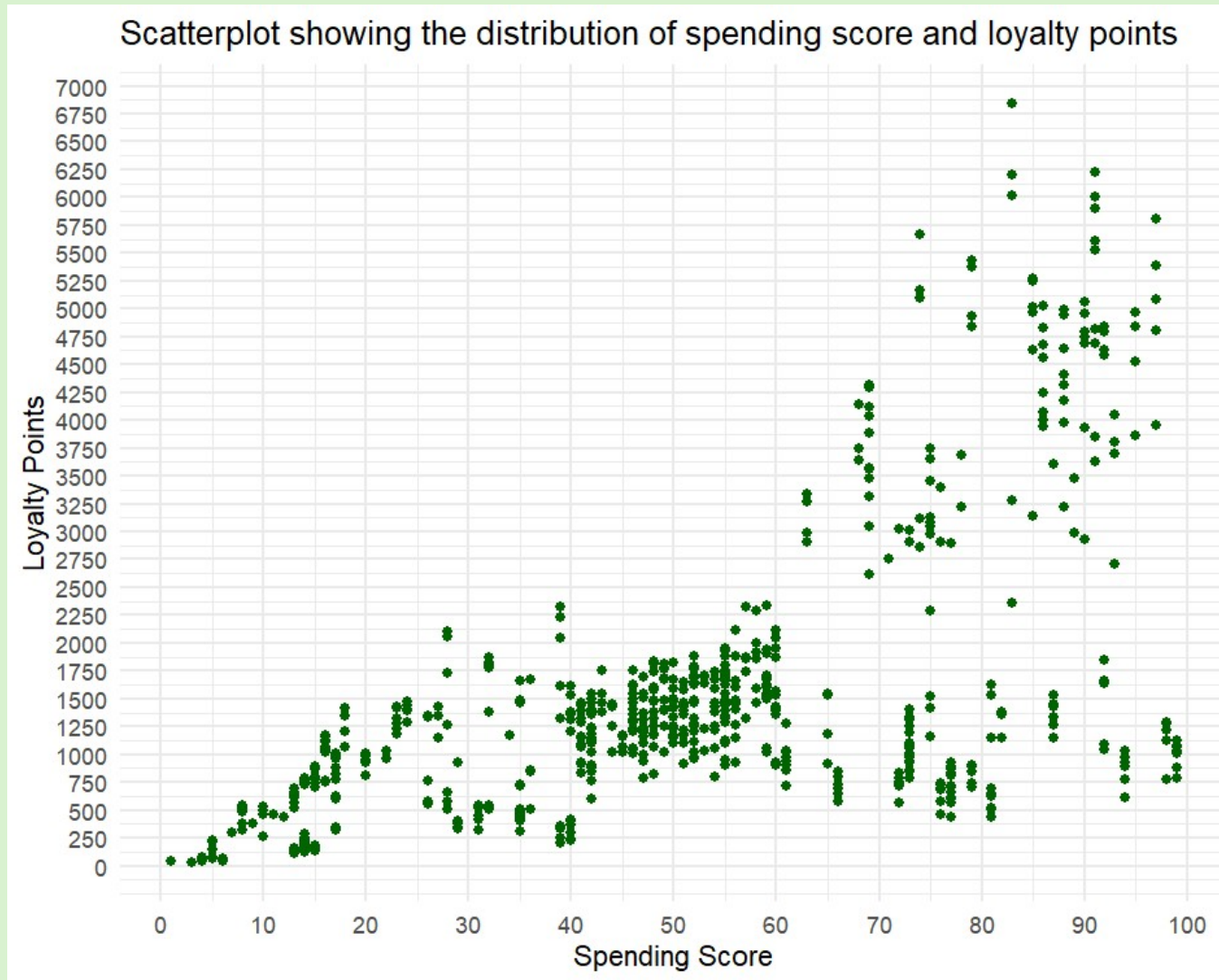
Scatterplot showing the distribution of income and loyalty points



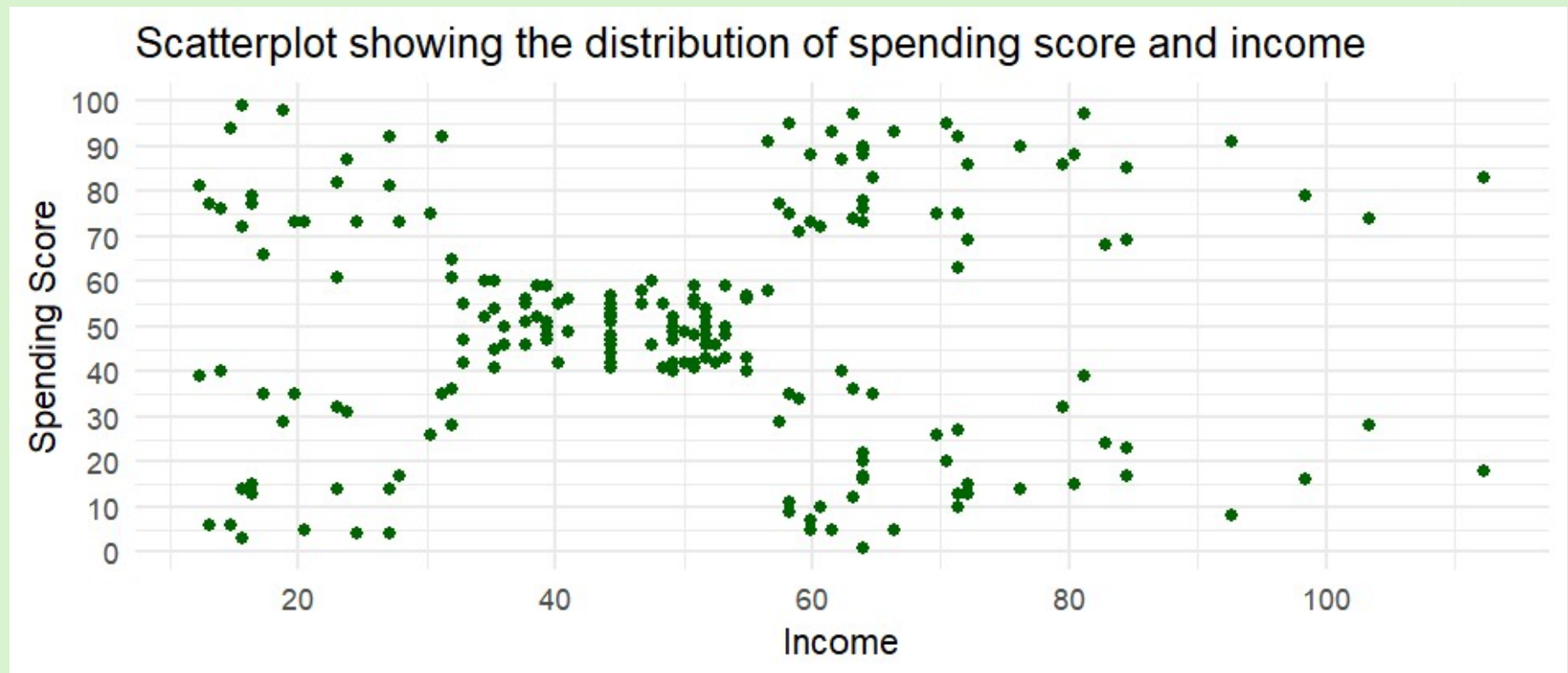
The higher the income, the higher the loyalty points, but not always



The higher the spending score, the higher the loyalty points, but not always



Can the data be grouped together?

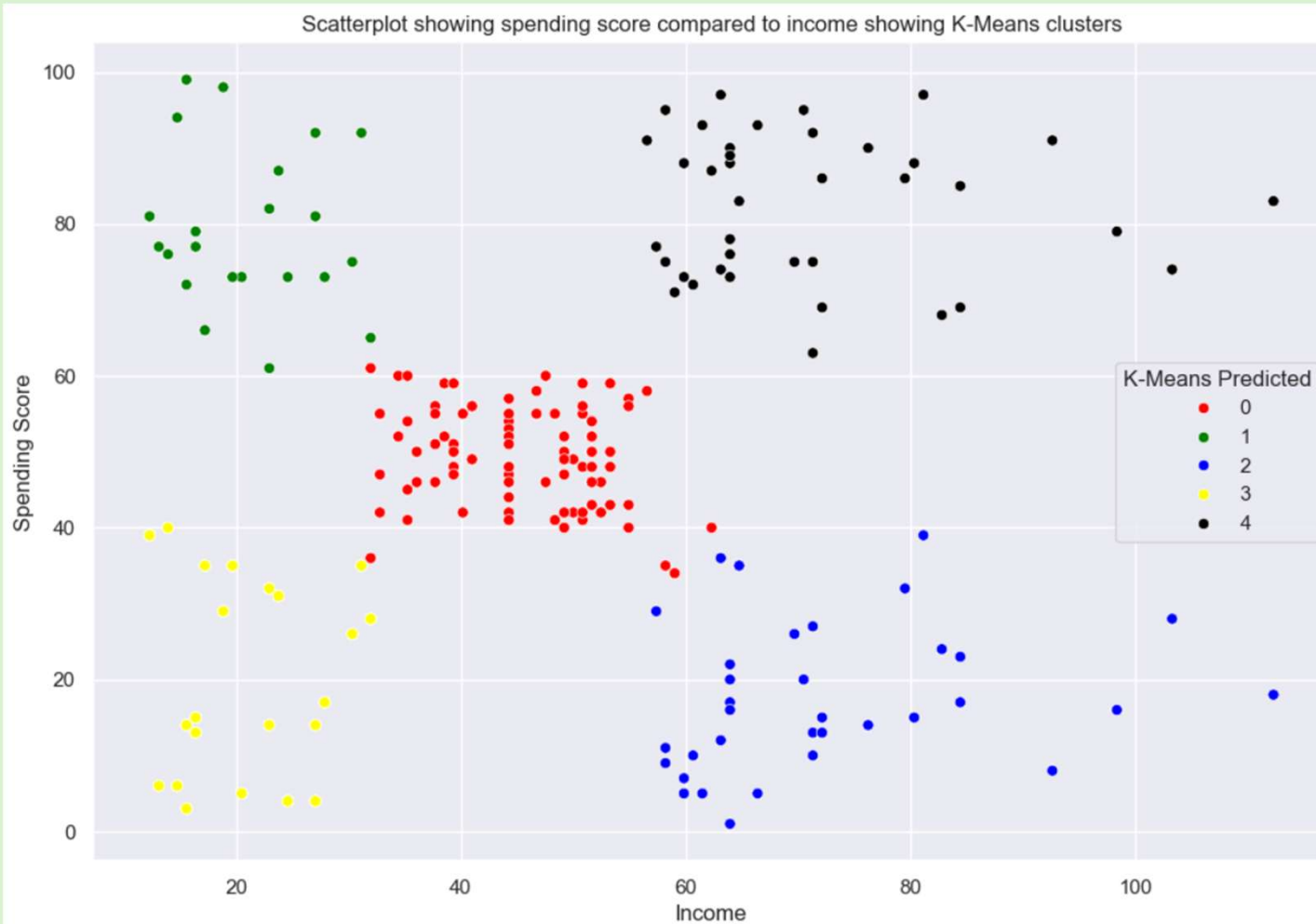




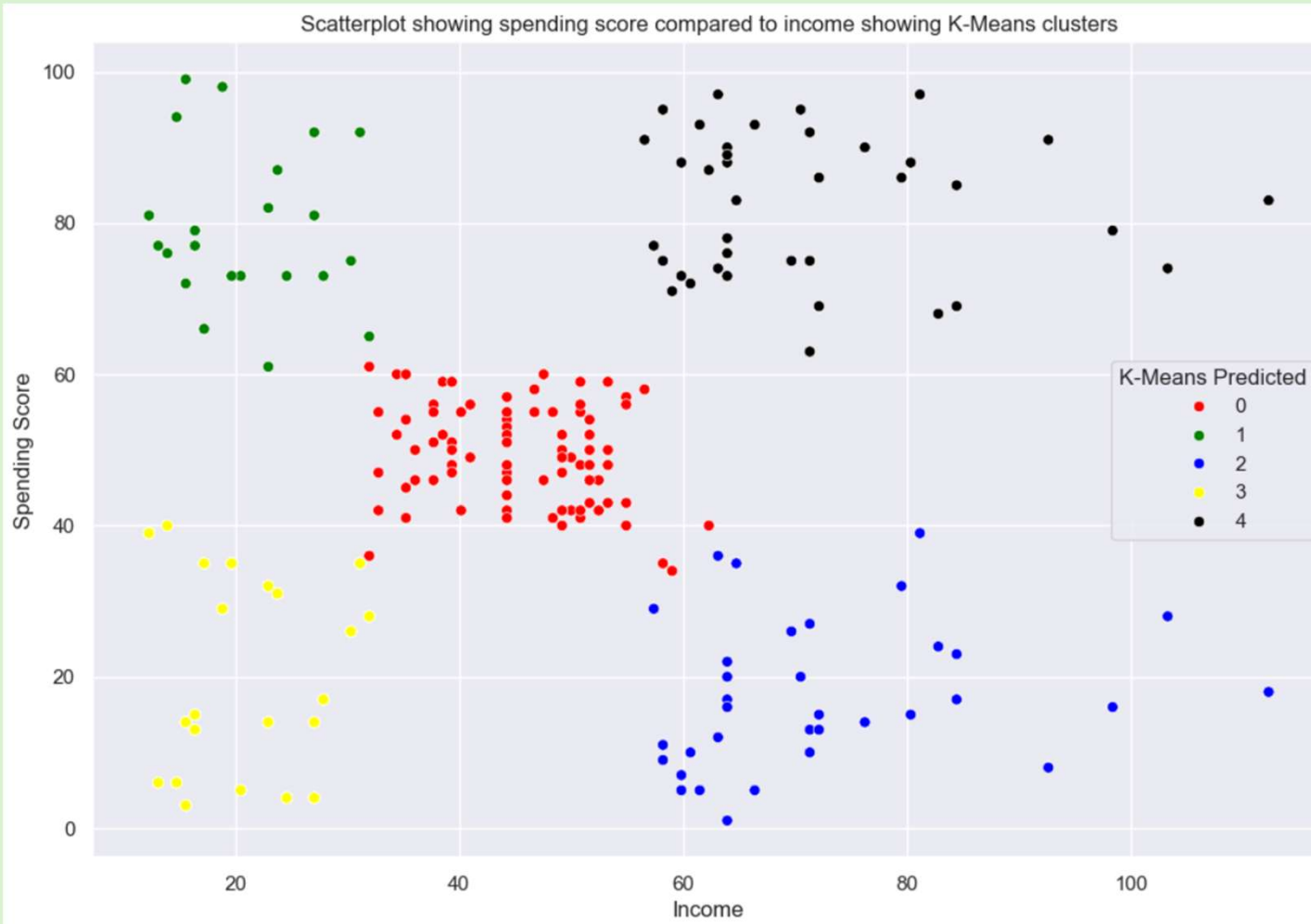
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The data
shows
five
distinct
groups



What do we know about each of these groups?



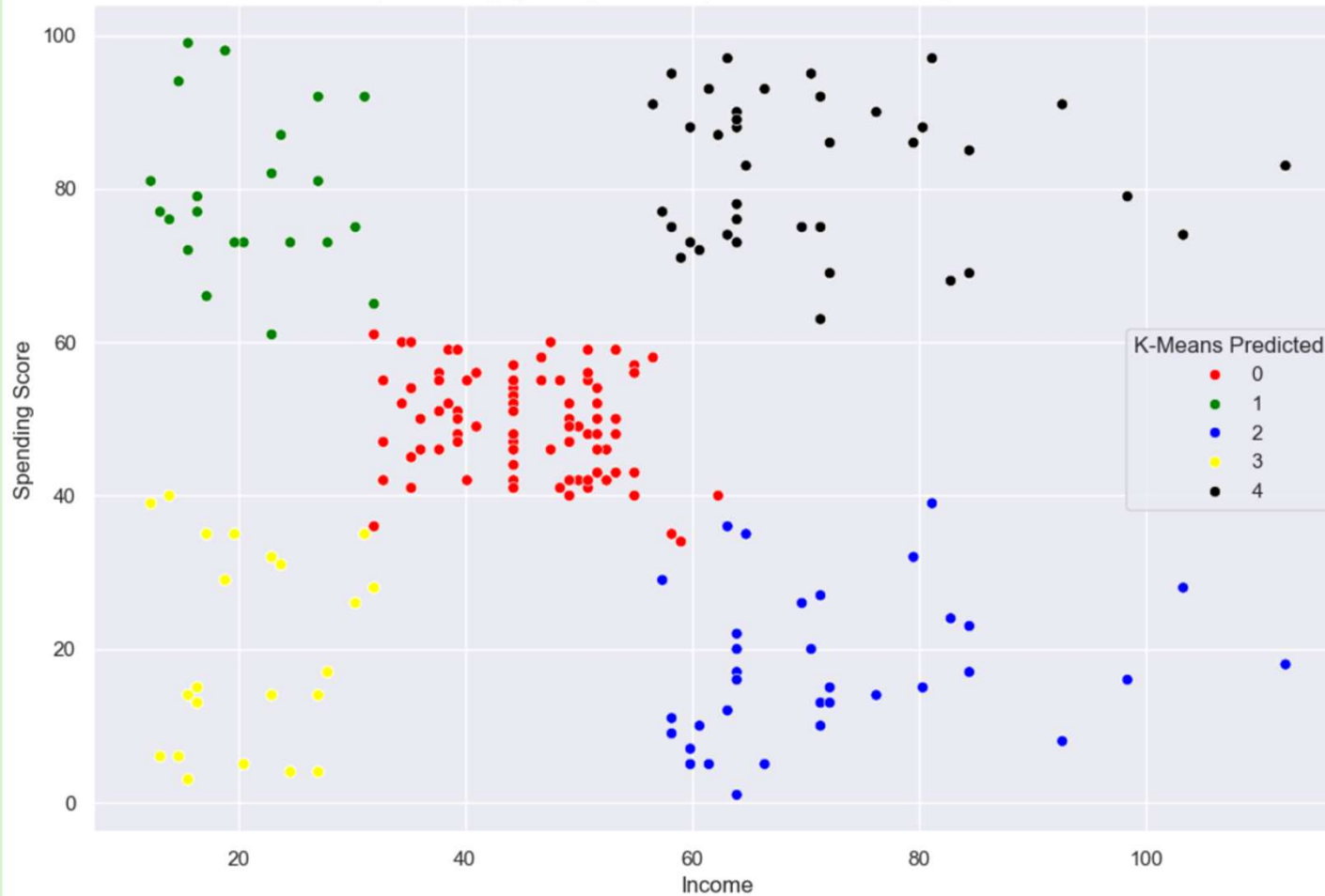
Average age – 42

Average income -
£44418.79

Average
spending score –
49

Average loyalty
points – 1420

Scatterplot showing spending score compared to income showing K-Means clusters





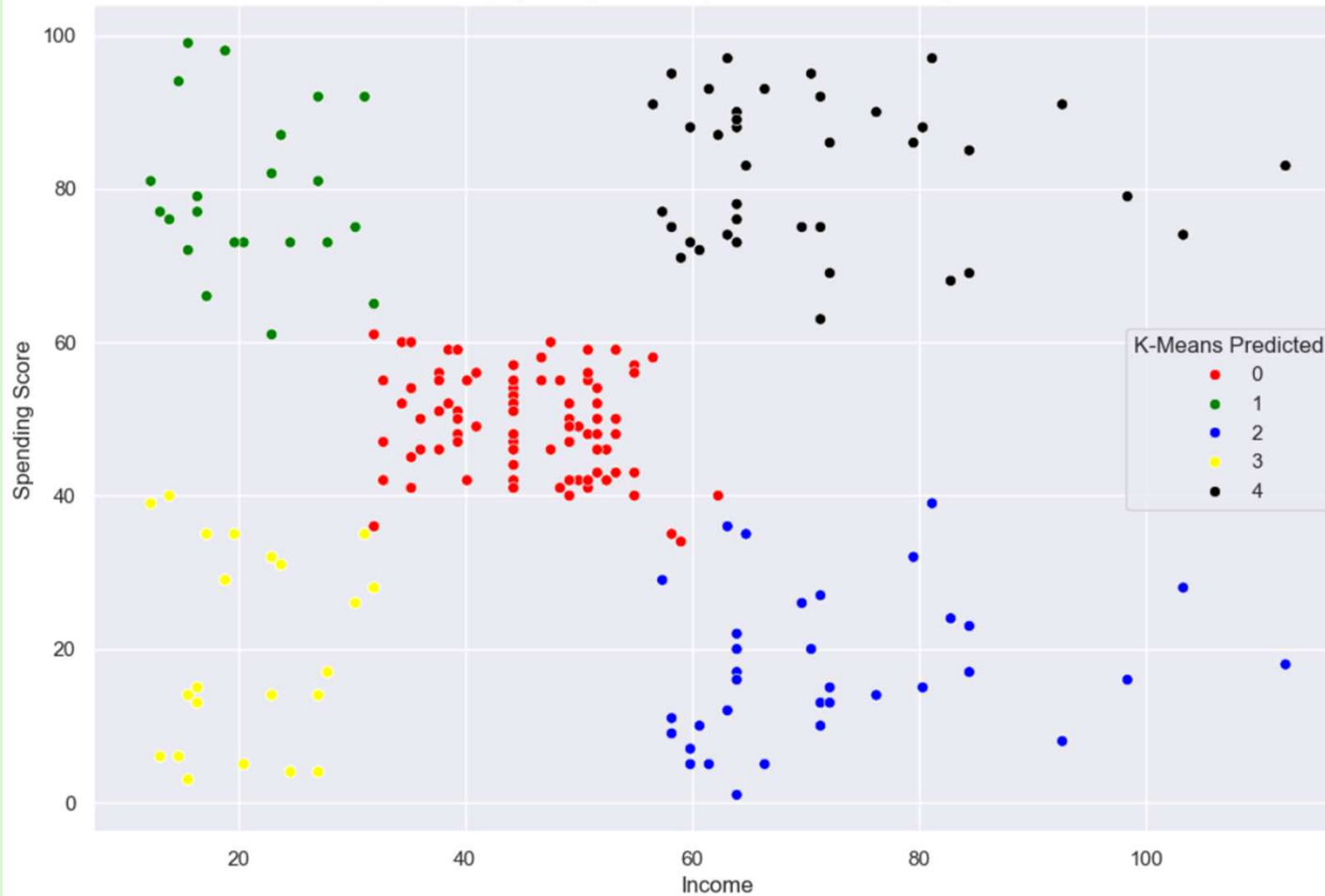
Average age – 31

Average income -
£20353.68

Average
spending score –
79

Average loyalty
points – 971

Scatterplot showing spending score compared to income showing K-Means clusters





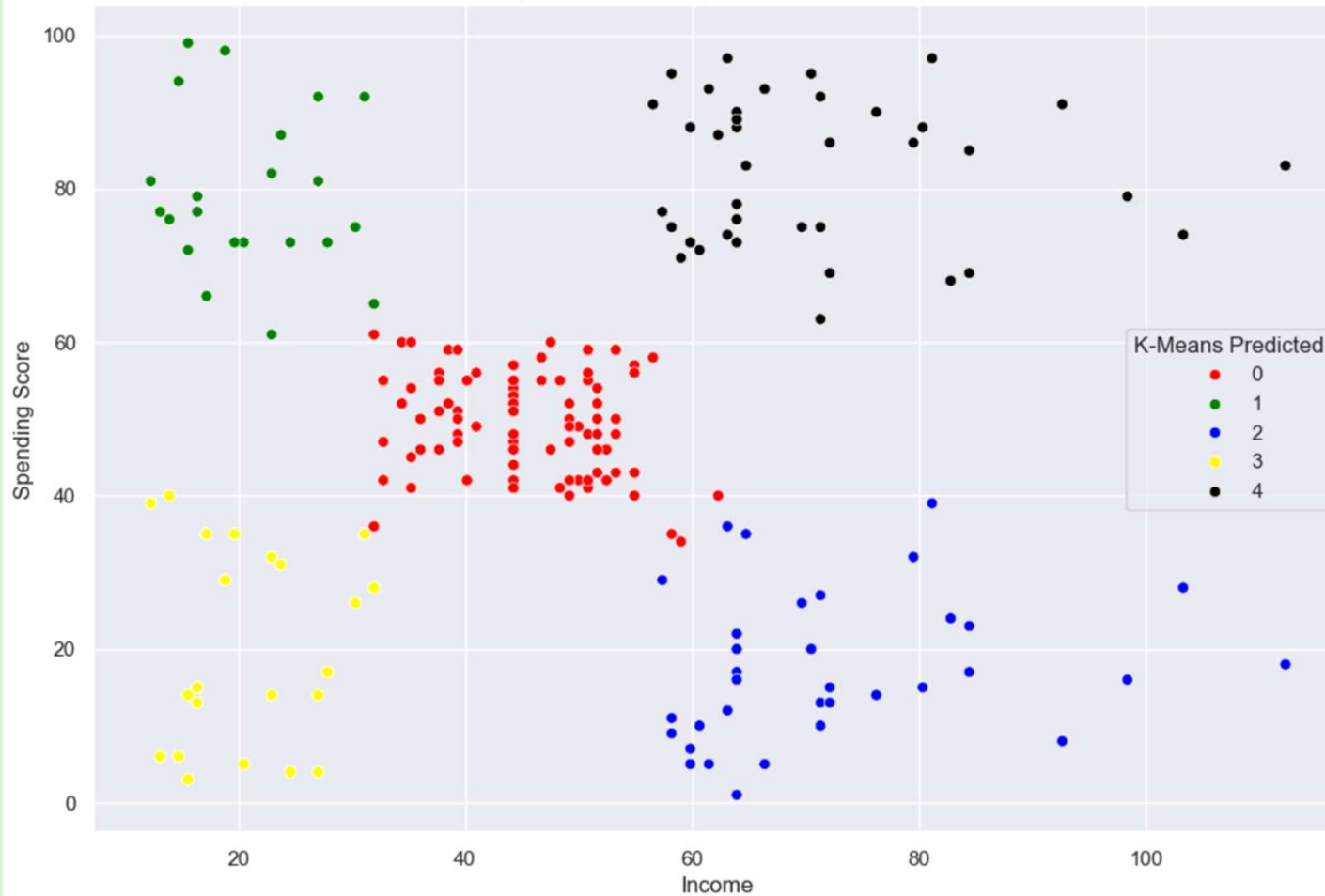
Average age – 40

Average income -
£74831.21

Average
spending score –
17

Average loyalty
points - 911

Scatterplot showing spending score compared to income showing K-Means clusters





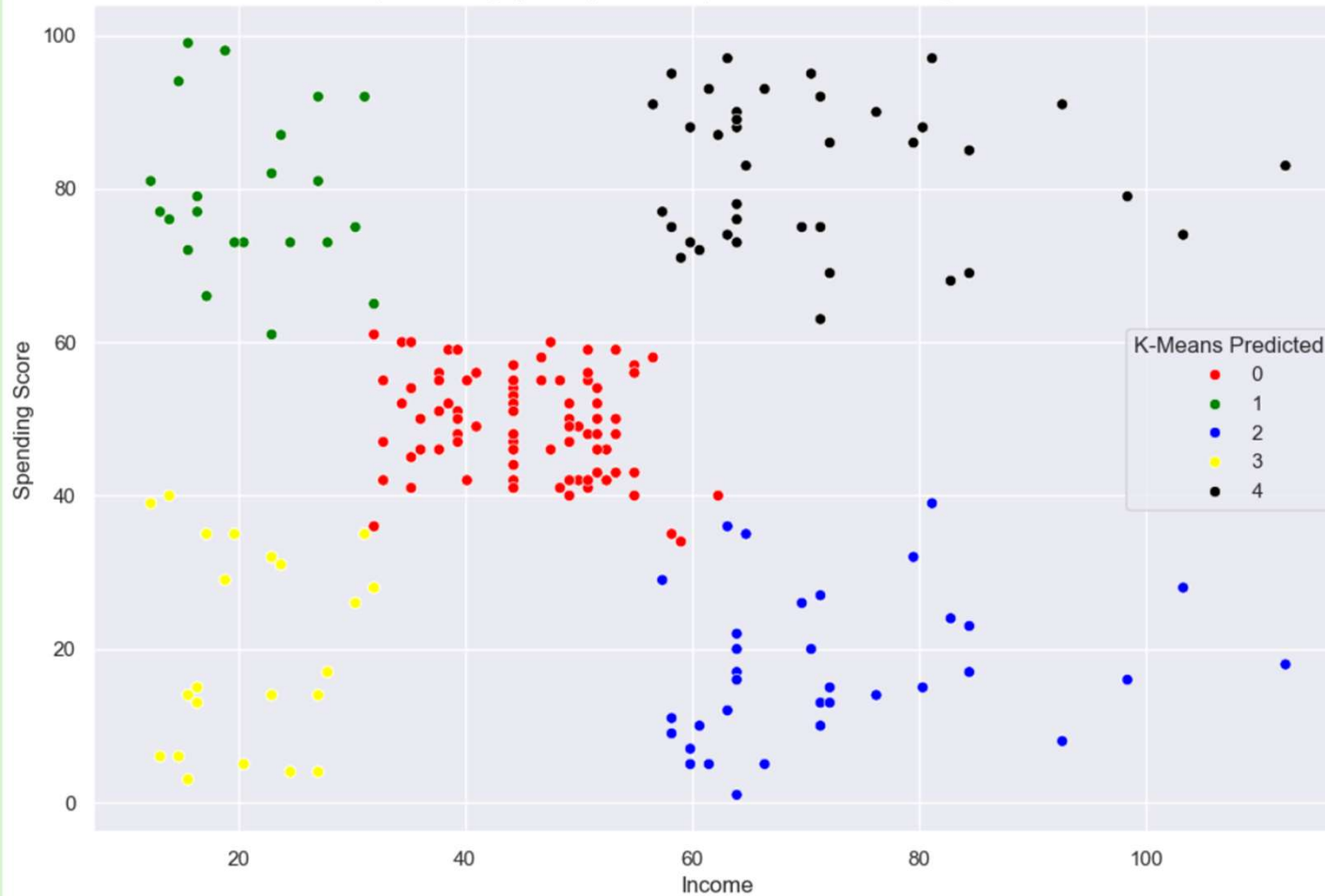
Average age – 43

Average income -
£20424.44

Average
spending score –
20

Average loyalty
points – 275

Scatterplot showing spending score compared to income showing K-Means clusters





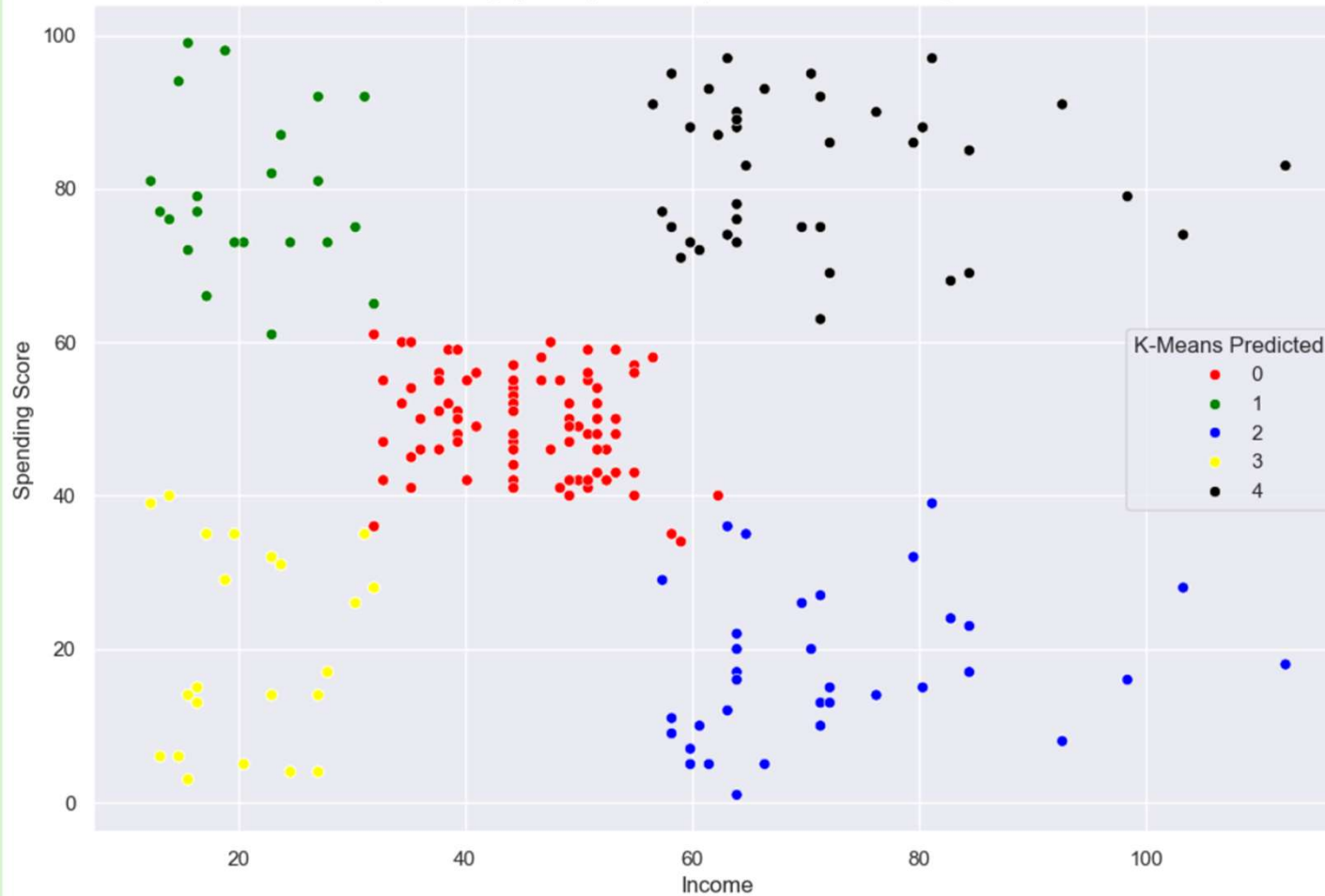
Average age – 36

Average income
£73240.28

Average
spending score –
82

Average loyalty
points – 3988

Scatterplot showing spending score compared to income showing K-Means clusters





Aims of the analysis

Turtle Games are aiming to improve overall sales performance by analysing and considering customer behaviour trends.

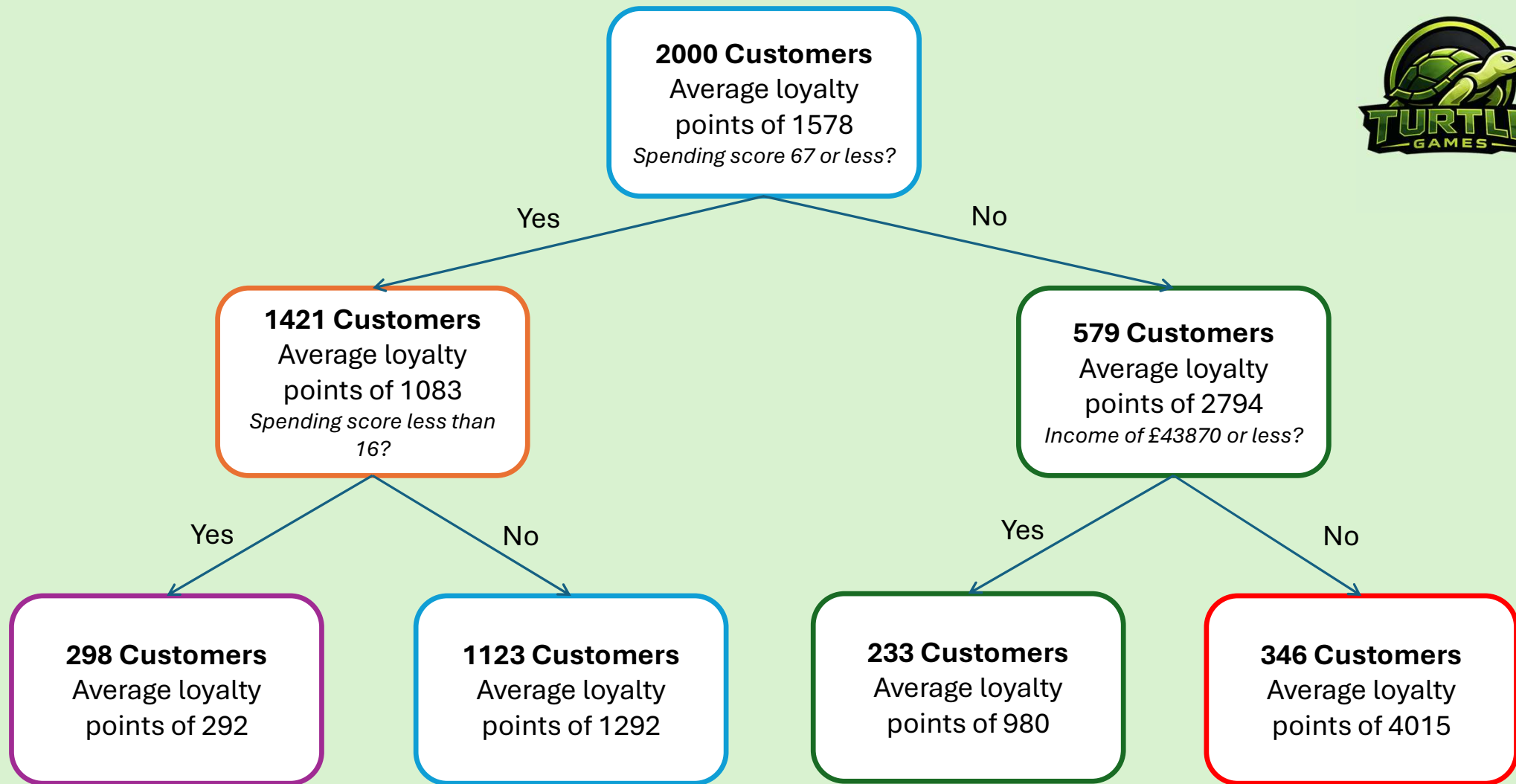
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This means that a spending score of 68 or more, and then an income greater than £43870 are important indicators for high loyalty points



Aims of the analysis

Turtle Games are aiming to improve overall sales performance by analysing and considering customer behaviour trends.

- How do customers engage with and accumulate loyalty points?



Aims of the analysis

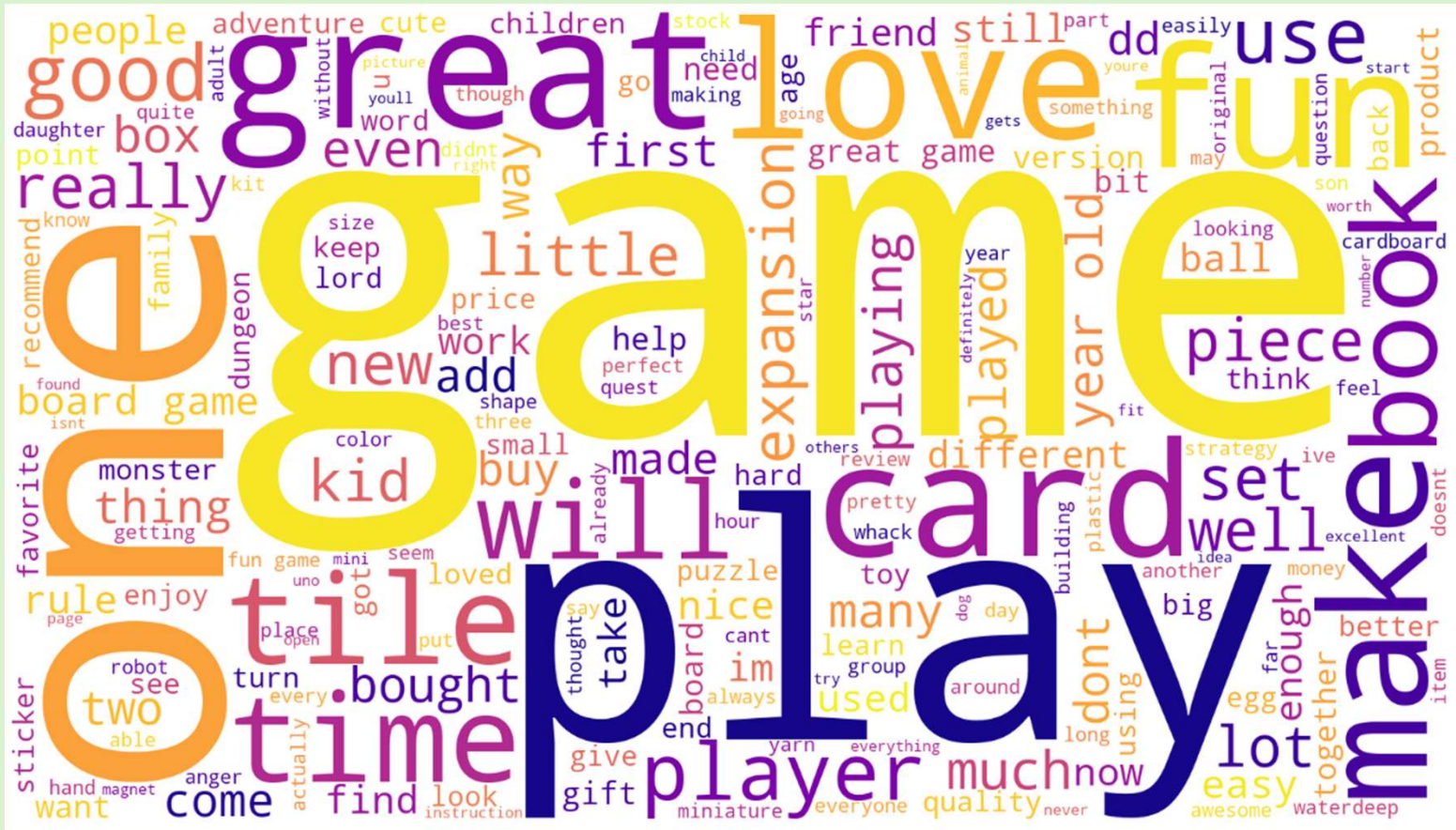
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What's in the data?

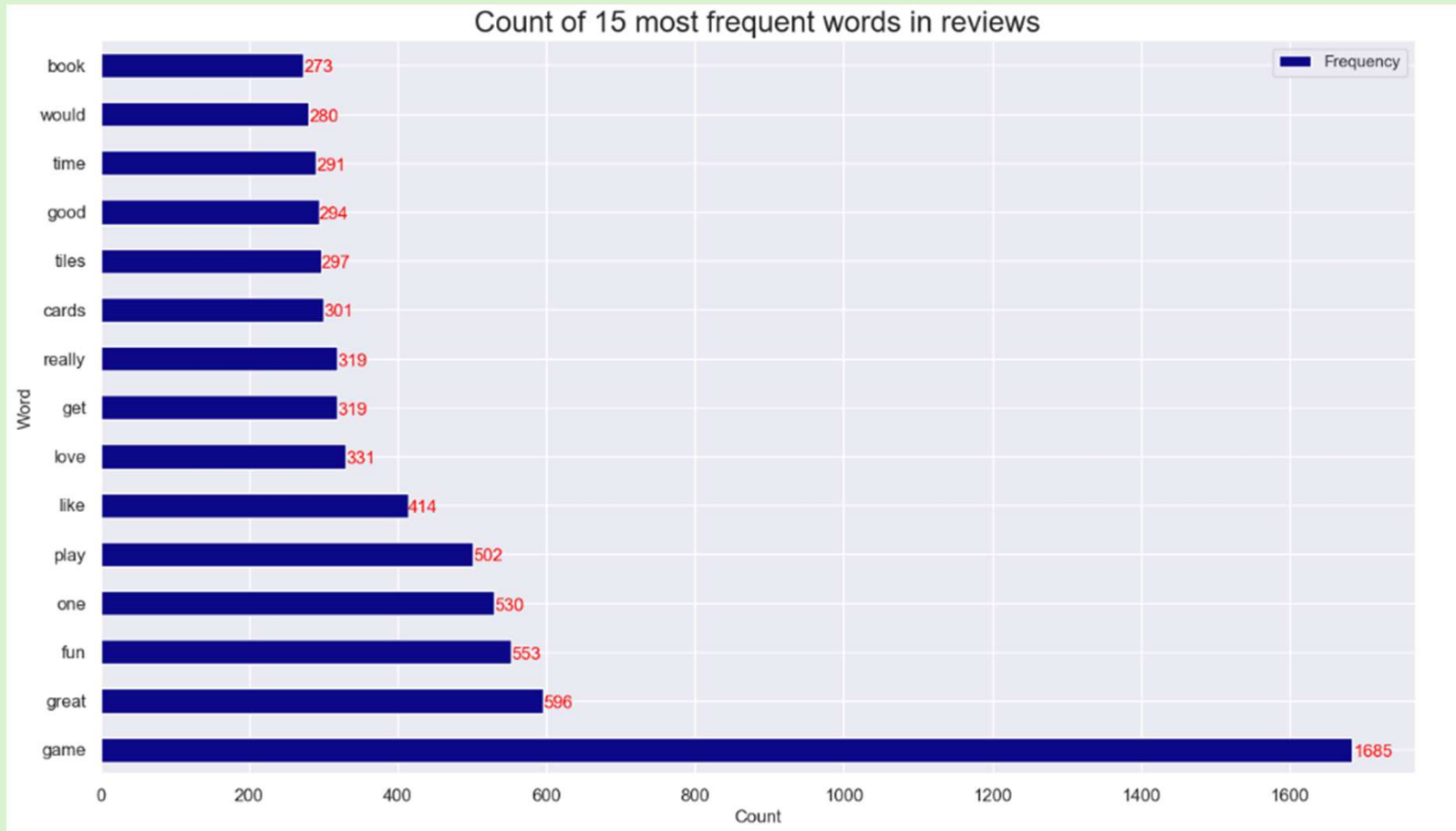
review	summary
When it comes to a DM's screen, the space on the screen its...	The fact that 50% of this space is wasted on art (and not ter...
An Open Letter to GaleForce9*: Your unpainted miniatures a...	Another worthless Dungeon Master's screen from GaleForce9
Nice art, nice printing. Why two panels are filled with a gen...	pretty, but also pretty useless
Amazing buy! Bought it as a gift for our new dm and it's per...	Five Stars
As my review of GF9's previous screens these were complet...	Money trap
Grandson loves	Five Stars
I have bought many gm screens over the years, but this one...	Best gm screen ever
Came in perfect condition.	Five Stars
Could be better but its still great. I love the things on the D...	Great but could be even better
My review will mirror others in that this kind of misses the ...	Another missed opportunity. Not a value add to the produc...
Works very well	Five Stars
Can't wait to use it!	Love the map!
This is a campaign specific DM screen meant to work with T...	Not a general DM screen
This is one of the worst games I have ever played. You basic...	Very weak game
It sounded like a really amazing concept, but when we tried ...	Fell completely flat...



Words in the reviews

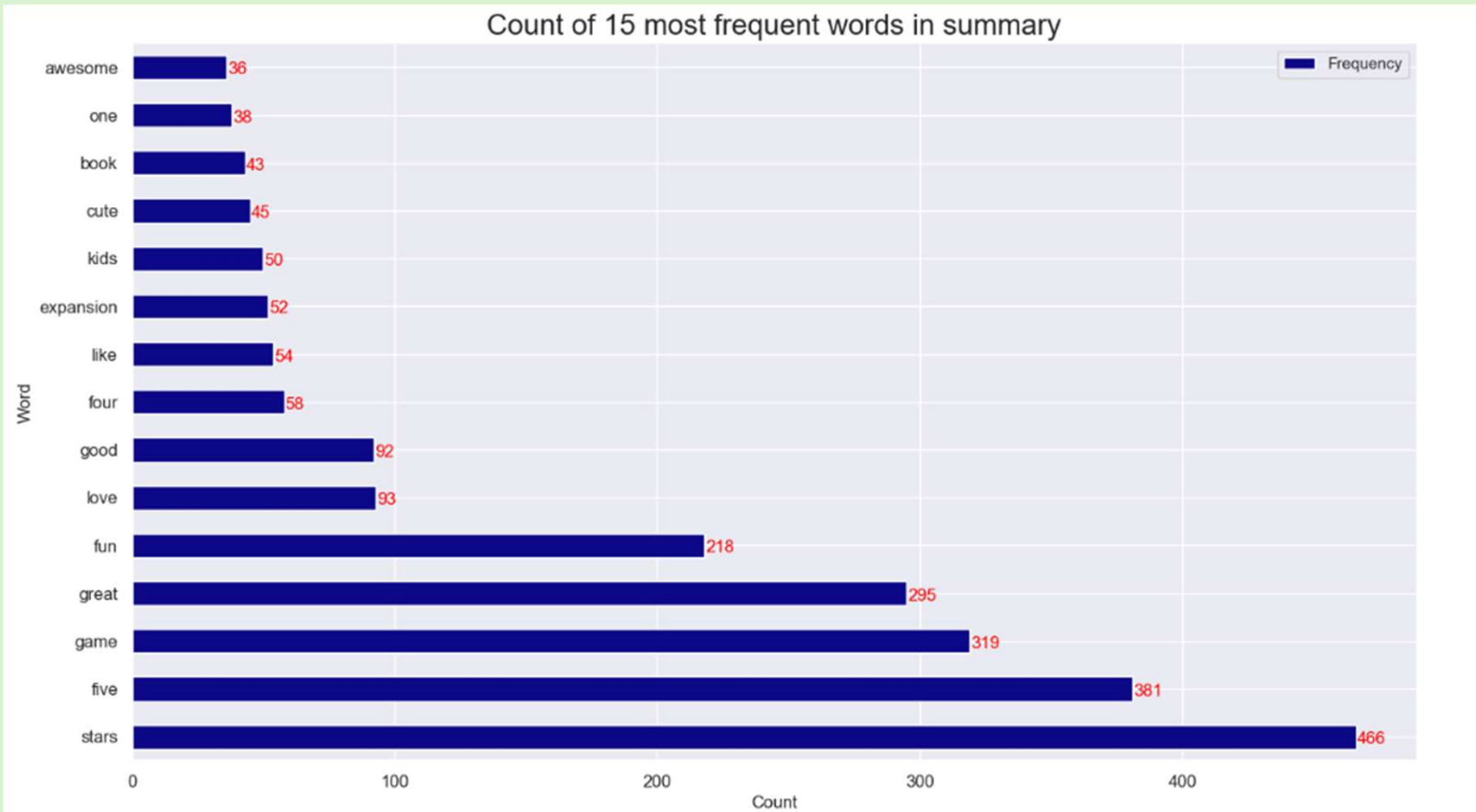


The most common word in the reviews is “game” along with “great” and “fun”



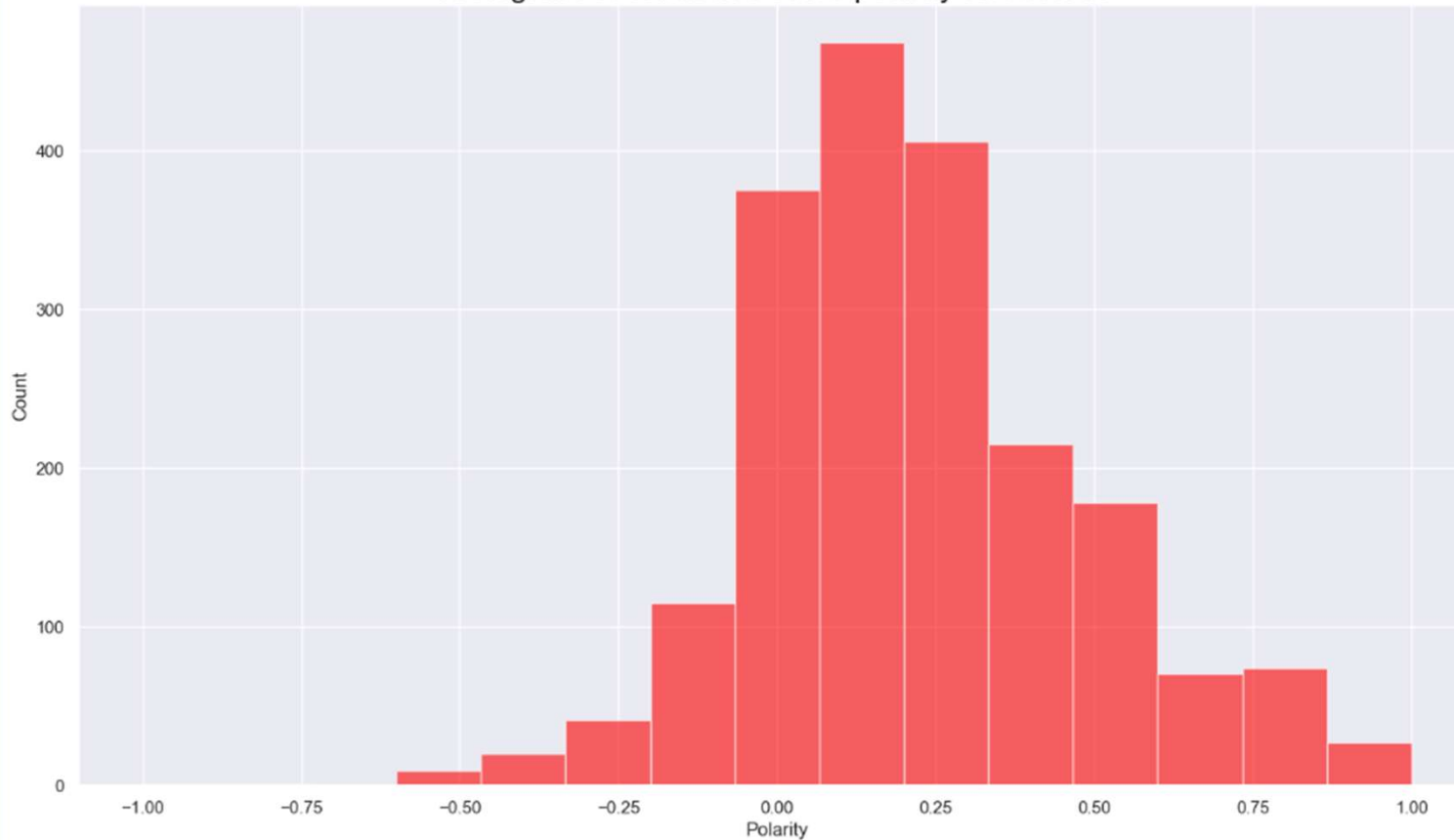


“Stars”,
“five” and
“four” are
amongst
the most
common
words in
the
summaries





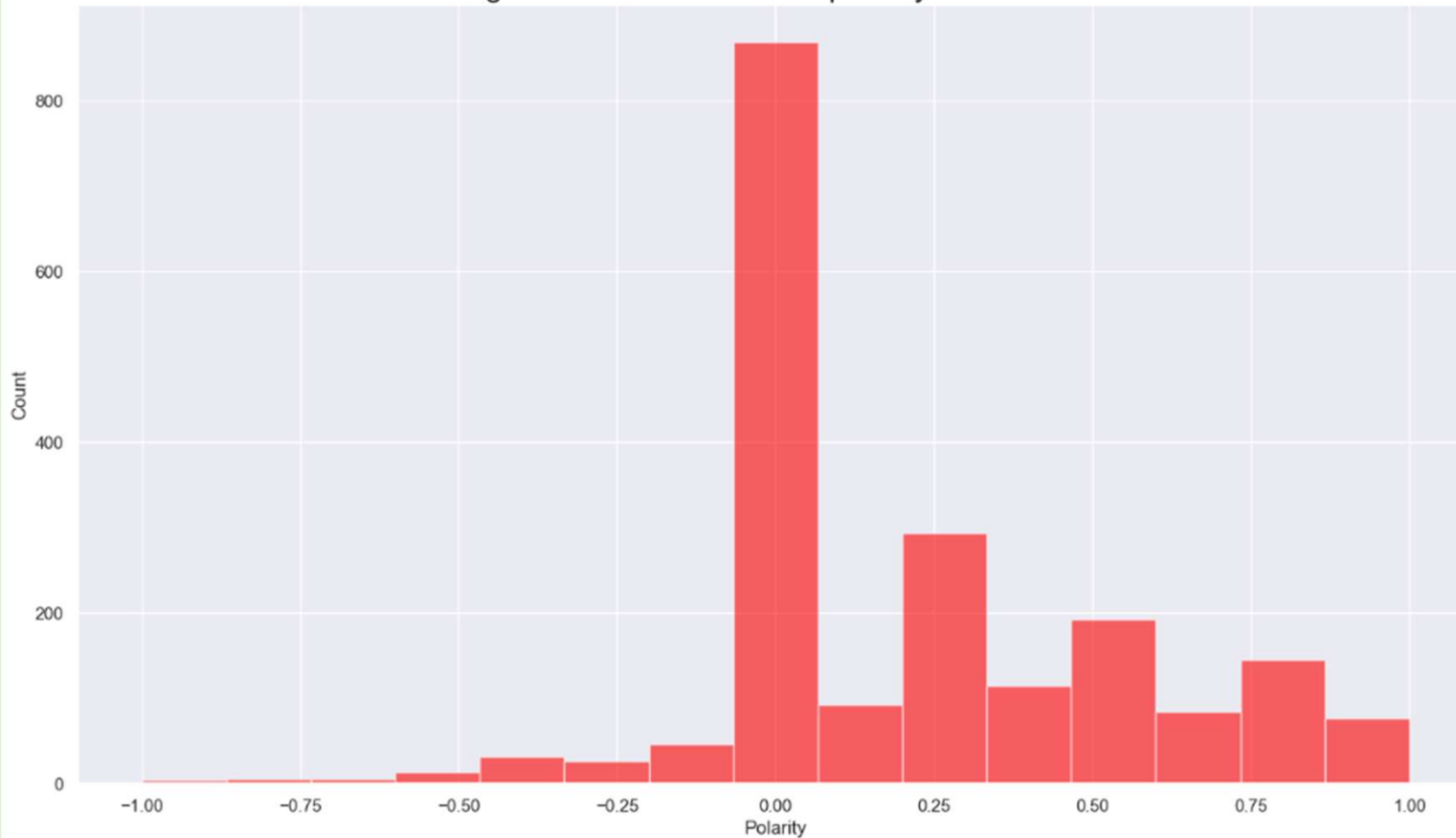
Histogram of sentiment score polarity for reviews



The
positive
reviews
outweigh
the
negative
reviews



Histogram of sentiment score polarity for summaries



The summaries are mostly neutral, but they lean towards positivity.



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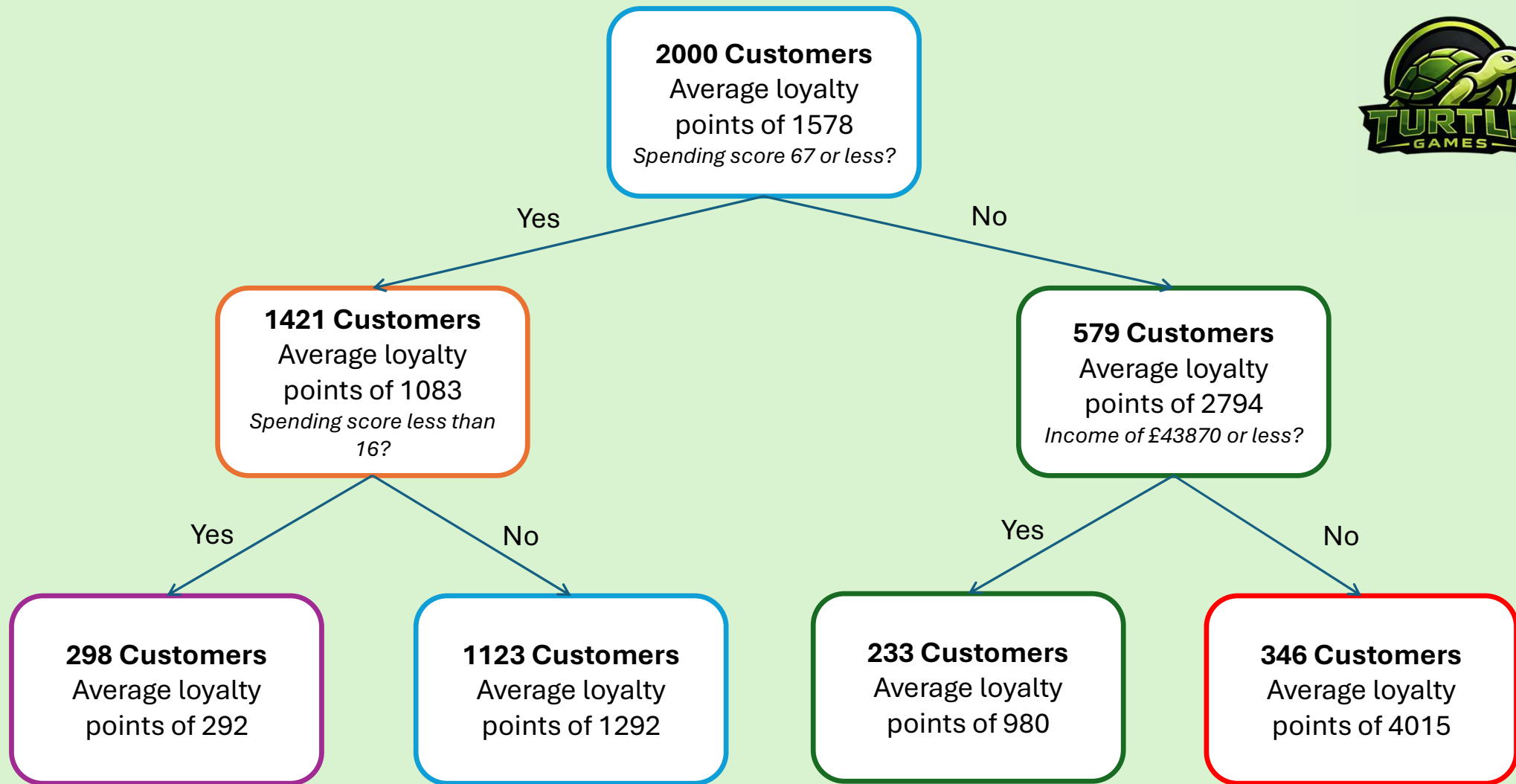
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How can we make predictions?

New Customer	Age	Spending Score	Income	Predicted Loyalty Points
New customer 1	45	50	40	1364
New customer 2	50	65	49	2238
New customer 3	55	70	43	2260
New customer 4	60	85	70	3747



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Recommendations

Looking at the groups of customers, Turtle Games should be aiming their marketing strategies towards the group with the high income, low spending score, and try and get that group to improve their spending score. Perhaps through target offers and deals that will entice that high income group to spend more money, and therefore accrue a lot of loyalty points. Ultimately benefit the company two-fold.

1. By getting the high income group to spend more money, that will benefit the company's overall sales performance, and therefore the profit will increase. More specifically if the high income group can improve their spending score by 50 points on average, this will create significantly high loyalty points, and benefit the company sales performance significantly
2. Getting more customers to obtain more loyalty points will encourage them to engage with the company more frequently, therefore improving the sales performance

Looking at a more generic marketing strategy, having demonstrably positive reviews will improve the company's standing with the wider public, and increase sales performance widely.



Recommendations

Looking at a more generic marketing strategy, having demonstrably positive reviews will improve the company's standing with the wider public, and increase sales performance widely.

Additionally, I recommend that Turtle Games should create a star rating for products when collecting data. This has been shown up in the summary column that the some of the most common words in the summary data shows four and five stars. This data will help enrich analyses in the future.



Thank you for watching